

# CSE400 – Fundamentals of Probability in Computing

## Lecture 11: Transformation of Random Variables

Lecture Scribe for Exam Revision

### Transformation of Random Variables

#### Lecture Objective

The lecture focuses on three connected goals:

- **Transformation of Random Variables**

Learn techniques to determine the distribution of a new random variable obtained from another.

- **Function of Two Random Variables**

Study transformations when the new variable depends on two random variables.

- **Illustrative Example**

Work through the case

$$Z = X + Y$$

These topics form the logical progression of the lecture:

single variable transformation  $\rightarrow$  two variable transformation  $\rightarrow$  example derivation.

### 1. Transformation of a Random Variable

#### Basic Setup

We start with a known random variable:

$$X \sim f_X(x)$$

Assumption:

The PDF of  $X$  is known a priori.

We define a new random variable

$$Y = g(X)$$

Goal:

Find the PDF of  $Y$ :

$$f_Y(y)$$

This means we must convert the probability description of  $X$  into the probability description of  $Y$ .

## Case 1: Monotonically Increasing Transformation

Assume:

$$Y = g(X)$$

and  $g(x)$  is monotonically increasing.

This assumption is important because it guarantees:

- the inverse function exists
- each  $y$  corresponds to exactly one  $x$

Thus

$$x = g^{-1}(y)$$

### Step 1 — Express the CDF of $Y$

Start from the definition of CDF:

$$F_Y(y) = P(Y \leq y)$$

Substitute  $Y = g(X)$ :

$$F_Y(y) = P(g(X) \leq y)$$

Since  $g$  is increasing, the inequality direction does not change, therefore

$$P(g(X) \leq y) = P(X \leq g^{-1}(y))$$

Thus

$$F_Y(y) = F_X(g^{-1}(y))$$

This step converts the probability into a form involving the known distribution of  $X$ .

### Step 2 — Differentiate to obtain the PDF

The PDF is the derivative of the CDF:

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

Substitute the CDF expression:

$$f_Y(y) = \frac{d}{dy} [F_X(g^{-1}(y))]$$

Apply the chain rule:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$$

Since

$$x = g^{-1}(y)$$

we can write

$$f_Y(y) = f_X(x) \frac{dx}{dy}$$

evaluated at

$$x = g^{-1}(y)$$

## Case 2: Monotonically Decreasing Transformation

Now assume  $g(x)$  is decreasing.

The key difference is the inequality direction flips.

### Step 1 — CDF expression

Start again from

$$F_Y(y) = P(Y \leq y)$$

Substitute transformation:

$$F_Y(y) = P(g(X) \leq y)$$

Because the function is decreasing:

$$g(X) \leq y \Rightarrow X \geq g^{-1}(y)$$

Thus

$$F_Y(y) = P(X \geq g^{-1}(y))$$

Using the complement rule:

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

### Step 2 — Differentiate

Differentiate the CDF:

$$f_Y(y) = \frac{d}{dy}[1 - F_X(g^{-1}(y))]$$

$$f_Y(y) = -f_X(g^{-1}(y)) \frac{d}{dy}g^{-1}(y)$$

The negative sign appears because the transformation is decreasing.

## Final General Formula

To combine both cases, the PDF is written using an absolute value:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

evaluated at

$$x = g^{-1}(y)$$

This formula ensures the PDF is always positive.

## Example: Transformation of a Uniform Random Variable

Given:

$$X \sim U(-1, 1)$$

So the PDF is

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Define the transformation:

$$Y = g(X) = \sin\left(\frac{\pi X}{2}\right)$$

Goal:

Find

$$f_Y(y)$$

### Step 1 — Find the inverse transformation

$$y = \sin\left(\frac{\pi x}{2}\right)$$

Solve for  $x$ :

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

### Step 2 — Compute derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

### Step 3 — Apply transformation formula

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute  $f_X(x) = \frac{1}{2}$ :

$$f_Y(y) = \frac{1}{2} \times \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Simplify:

$$f_Y(y) = \frac{1}{\pi \sqrt{1-y^2}}$$

## Step 4 — Determine support

From the transformation:

if  $x = -1$

$$y = \sin(-\pi/2) = -1$$

if  $x = 1$

$$y = \sin(\pi/2) = 1$$

Thus

$$-1 < y < 1$$

Final PDF:

$$f_Y(y) = \begin{cases} \frac{1}{\pi\sqrt{1-y^2}}, & -1 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

## Function of Two Random Variables

The lecture next considers the transformation

$$Z = X + Y$$

Goal:

Find the PDF

$$f_Z(z)$$

## Deriving the Distribution of $Z = X + Y$

Start with the CDF definition:

$$F_Z(z) = P(Z \leq z)$$

Substitute the definition of  $Z$ :

$$F_Z(z) = P(X + Y \leq z)$$

This probability corresponds to a region in the  $x$ - $y$  plane.

To compute it we integrate the joint PDF over that region.

### Horizontal strip integration

$$F_Z(z) = \int_{y=-\infty}^{\infty} \int_{x=-\infty}^{z-y} f_{XY}(x, y) dx dy$$

inner limit:  $x \leq z - y$  because  $x + y \leq z$

### Vertical strip integration

$$F_Z(z) = \int_{x=-\infty}^{\infty} \int_{y=-\infty}^{z-x} f_{XY}(x, y) dy dx$$

Both integrals describe the same probability region.

## Special Cases Mentioned

Find  $f_Z(z)$  in general.

If  $X$  and  $Y$  are independent.

If

$$X \sim N(0, 1), \quad Y \sim N(0, 1)$$

then

$$Z = X + Y$$

and it can be shown

$$Z \sim N(0, 2)$$

If  $X$  and  $Y$  are exponential random variables with parameter  $\lambda$ , compute  $f_Z(z)$ .

## Key Logical Flow of the Lecture

- Start with known distribution of  $X$ .
- Define a new random variable  $Y = g(X)$ .
- Express CDF of  $Y$  using probabilities of  $X$ .
- Differentiate to obtain the PDF of  $Y$ .
- Handle increasing vs decreasing functions.
- Apply the formula to a worked example.
- Extend the concept to two random variables.
- Use joint PDF integration to compute distributions like  $Z = X + Y$ .